

SEDAR 3: South Atlantic/Gulf of Mexico Yellowtail Snapper and ASMFC Atlantic Croaker and Menhaden

South Atlantic/Gulf of Mexico Yellowtail Snapper

Assessment Workshop

- As with other fisheries, we need data on all removals from the fishery.
- We need to collect annual discard information from all sectors of the fishery.
- An improvement for the assessment would be to develop a probabilistic aging procedure that accounts for selectivity and mortality that uses the catch-at-length and fishery-independent and fishery-dependent ages and lengths.
- We need to investigate the inclusion of interaction terms in the calculation of standardized catch rates.
- We also need to investigate whether the increases in the commercial catch rates reflects improvements in fishing methods such that the increase does not reflect the underlying population.
- We also need to review the methodology of the Reef Visual Census and its use as a fishery independent index of population trends.
- Another catch rate issue is whether the change in contractors for MRFSS was responsible for the patterns in the recreational catch rates.
- Stock assessments in the Southeastern U. S. would benefit from a workshop addressing natural mortality and steepness and how the stock status conclusions depend on the chosen values.
- The performance of the assessment models could be evaluated for retrospective bias by running the models with simulated data.

Review Workshop

Yellowtail Snapper Specific Recommendations

- Determine the release mortality rate for fish in the commercial, charterboat, headboat, and private/rental boat fisheries.

- Collect discard data (quantity, size, condition, etc.) from the headboat fishery. This could include modification to the current logbook used by headboats or employing observers; if observers are used, they could also collect biological data. Collection of discard data from the commercial fishery should continue. It is critical that a total (accurate) estimate of discards by sector (commercial, headboat, charter boat and private/rental boat) be available for the next assessment.
- Thoroughly evaluate the reef visual census CPUE index prior to use in future assessments.
- Examine alternative methods to incorporate recent increases in catching efficiency (“power-chumming”, smaller hooks, fluorocarbon leaders, GPS, etc.) into the commercial and recreational CPUE indices. This effort should lead to alternative methods to refine CPUE indices (electronic logbooks, observers, etc.), or alternative indices.
- Continue the use of annual age/length keys, and move to direct age estimation where possible. Cognizance should also be taken of the temporal and geographic effects on such collections.
- Seek better validation of age estimates.
- Continue research into stock structure, e.g. genetics.

General Assessment & Research

- Thoroughly examine estimates of natural mortality (M) and steepness (h) in a workshop setting. Such a workshop should not be limited to yellowtail snapper, but should make comparisons with other species.
- Examine the following issues with the MRFSS program:
 - * The contractor changed in the mid-1990s. Whether or not this affected CPUE trends should be examined.
 - * The level of intercepts increased after 1992, and from 1998/99 onwards, representatives of the State of Florida conducted the intercepts. What impact did this have on estimates and how should this CPUE index be incorporated into future assessments (as a continuous time-series or subdivided into one or more separate time series)?
 - * Private vessel owners leaving from their own private facilities are not currently sampled adequately. Is an adjustment factor used to account for this sector? Is this an important issue in Keys fisheries, given the large number of canals and private docks?

- * Given the concerns about the MRFSS data, potential new methodologies to collect these data should be evaluated.
- Examine predator/prey interactions (and other ecosystem considerations).
- Develop methods to incorporate the effects of spatial variability into assessments.
- Put effort into developing better fishery-independent survey indices to assess fish stock status.

Comments Regarding Goliath Grouper

Goliath Grouper were initially considered during the data workshop but further assessment effort was not pursued due to a lack of data. The Review Workshop identified some potential assessment methods and overlooked datasets that ultimately led to an assessment of Goliath Grouper that was reviewed through SEDAR 6.

- Estimation of population size. Estimates of population size were considered to be crucial for future management. It was noted that, because of the apparently narrow home ranges and site fidelity, sampling throughout the geographic range would probably be important. Tag/recapture research and studies with data storage tags were mentioned as potential monitoring tools.
- Demographics. Monitoring the demographics of the population, particularly age composition, could provide valuable information. Noting that age determination of the species was difficult, the Panel suggested that effort be channeled into improving it.
- Reproductive biology. Developing further understanding of the reproductive biology of goliath grouper was considered important. Identifying spawning locations, duration and periodicity, and identifying whether there were spawning migrations, could be useful in identifying sites to conduct population surveys. Further, there would be value in obtaining more information on early life history (eggs and larvae). It appeared that the survival rate of juveniles in mangroves and estuaries was good.
- Historical abundance. Obtaining information on historical abundance, perhaps via old logbooks, was considered a possibility as such information could enhance assessments.
- Other research material and topics considered as of less immediate importance or of questionable feasibility (in terms of collection of data) were:
 - estimating unrecorded mortality from accidental or intentional sources;
 - information on stock structure;

- bioenergetics and trophic relationships (though note the comment above on the need for ecosystem management);
- information identifying changes in mangrove abundance and distribution, and hence changing available nursery habitat (goliath grouper spend their first 6-7 years in mangrove areas, sometimes attaining as much as 50 lbs).

Process and Procedure Recommendations

- Provide hard copies of materials for participants. Not everyone can access material via the Internet and download/print large quantities of material.
- The category “recreational catch” should include charterboat catches, private/rental boat catches, headboat catches and shore-mode catches (if appropriate).
- Review and evaluation of data during Data Workshops should be much more rigorous. All data should be plotted and the trends examined, and detailed recommendations should be documented and provided on the use of the various datasets. Assessment scientists should attend along with representatives of all major data collection programs (MRFSS, commercial logbook, TIP, etc.). Consensus needs to be reached on the use of specific datasets or estimates for incorporation in the assessments.
- The next assessments should use simple stock assessment techniques in addition to relatively complex stock assessment models, because simple techniques are easier to understand and describe, as well as being useful in confirming the results from the more complicated models. In particular, simple exploitation indices (total catch divided by abundance indices) should be examined to detect trends in fishing mortality. The simple trends in survey, CPUE, and catch data should be examined and described, and trends in survey and CPUE data compared. Trends in mean length or mean weight also provide information on exploitation and recruitment levels, and are worthy of presentation.

CIE Consultant Recommendations

- That consideration be given in future assessments to:
 - * the issues of year interactions, polynomial terms, and model selection in the standardization of CPUE ;
 - * the use of less arbitrary data weightings;
 - * further validation of yellowtail snapper ageing, an examination of the “representativeness” of age-length keys, and more work on direct age estimation;

- * better documentation of the rationale for the assumed values of natural mortality and recruitment steepness;
- That consideration be given, in writing of terms of reference for future SEDAR Assessment Review Panels, to
- * either removing the phrase “including management recommendations” or giving clear guidance as to what sort of management recommendations are appropriate;
- * clarifying what is to be reviewed — the assessment or the assessment report — and, if the latter (not recommended), providing clear guidelines as to what is required in an assessment report.

Atlantic Menhaden

- There is no adult abundance index to tune the population model.
- * Evaluate commercial purse seine fishery effort (vessel/weeks) series as a possible tuning
- * index in the model. Evaluate any measure of effort contained in this or other data series.
- * Evaluate the data collected in the Captain’s Daily Fishing reports for an adult abundance
- * index. If these data are not useful, explore the utility of a commercial fishery-based adult
- * index, developed jointly with the fishermen, for future assessments.
- Recent relative productivities of menhaden nursery areas coast wide are unknown.
- * Investigate if there are any existing studies that could assist in evaluating current productivity.
- * Develop protocols to quantify contribution of different nursery areas to the adult stock.
- M-at-age is an improvement over constant M assumption. However, there is concern that not all key sources of mortality have been accounted for and little is known about the temporal patterns of mortality.
- * Identify key sources of non-fishing mortality for menhaden.
- * Enhance the coverage of the MSVPA to more predator and prey species.
- * Determine if there are temporal patterns in these sources.
- * Validate assumptions about applying results from MSVPA to the 1955-1980 period.

- There have been large changes in size-at-age over the 1955-2002 period. These trends are not a problem for the model but could have an impact on forecasts.
- * Evaluate historical change in size (weight and length) at age using existing data (e.g., scale incremental widths).
- There are patterns in residuals of numbers at age for commercial catch estimated by the model.
- * Investigate if the selectivity model is causing this pattern.
- * Look at spatial changes in fishing pattern as well as fish distribution.
- Current fecundity estimates are from studies in the 1980's and earlier.
- * Update the fecundity-at-size estimates and maturity ogives.
- Cannot address local depletion questions with the current model.
- * Investigate methods to determine the proportion of the stock that may reside in a particular area in any one season and whether regional reference points can be developed to address local depletion.
- * Extend these methods to track changes in distribution over time.
- Control plot determination of overfishing/overfished is based on point estimates only.
- * Develop uncertainty measures or risk analysis for control plots.
- It is difficult to distinguish between results of different models and model assumptions.
- * Develop measures (goodness of fit/complexity) to screen multiple models.
- The assessment model assumes a unit stock.
- * Test this assumption using otolith microchemistry and/or genetic markers.

Atlantic Croaker

The Review Panel rejected the initial Atlantic Croaker stock assessment due to critical data and model deficiencies. Specific steps necessary to correct the assessment were outlined as well as long term research and monitoring needs.

Recommendations to correct initial model

- Commercial landings did not include all removals from the population.

- * Evaluate North Carolina uncultured bait (“scrap”) fishery data and include in the commercial landings.
- * Evaluate the potential of applying the North Carolina uncultured bait fishery data to other states.
- * Consider at-sea observer data for discards and bycatch.
- The model used catch data from 1973 to the present but tuning indices were only used from 1981 to the present.
- * Extend the NMFS NEFSC bottom trawl survey data to 1973 for inclusion in the model.
- * Evaluate the difference between the Delta lognormal and stratified mean estimates from NMFS NEFSC bottom trawl survey.
- * Evaluate the VIMS survey data for possible inclusion in the model.
- The base model assumed that the SSB in 1973 was equal to 0.75 SSB (virgin biomass) from the Beverton-Holt analysis.
- * Re-evaluate after inclusion of the full time series of NMFS NEFSC and VIMS trawl survey data.
- The model assumes that the fisheries-independent survey indices are more precise than the fisheries-dependent data and model recruitment estimates and, therefore, provided higher weights to these surveys.
- * Evaluate the consequences of alternative weighting schemes.
- * Provide detailed justification for the final choice of weighting scheme.
- Separate models were developed for the mid-Atlantic (North Carolina and north) and South Atlantic (South Carolina to Florida).
- * Investigate the distribution and movement of croaker by age and season.
- * Compare life history parameters over the full distribution of croaker.
- The assessment included an age structured production model only. This required development of an algorithm to generate an age structure for the population.
- * Compare non-age assessment models, such as the Collie-Sissenwine catch-survey and a delay difference model, to understand the implications of this age structure on derived reference points and stock advice.

- Determination of overfishing/overfished were based on point estimates only.
- * Estimate the error distribution for current estimates of F , and reference points.
- * Determine whether, given error distributions determined above, target F and threshold F could be distinguished from estimates derived from the assessment model.
- * Consider revising F target reference point relative to the previous bullet.

Research Recommendations

- Separate models were developed for the mid-Atlantic (North Carolina and north) and South Atlantic (South Carolina to Florida).
- * Conduct tagging and otolith microchemistry studies to address the justification for regional assessments.
- Difficult to understand what component of the population the surveys were tracking.
- * Include maps of fishery and survey areas in future reports.
- A single growth curve based on data from North Carolina was applied over all years and for whole area.
- * Evaluate the applicability of the North Carolina growth curve to all areas (spatial variability).
- * Investigate inter-annual variability in growth.
- A single natural mortality estimate was used for all ages and years.
- * Develop age-specific M for inclusion in the model.
- Trends in the recruitment deviations may indicate temporal bias in the recruitment model.
- * Assess whether changes in potential population reproductive capacities have changed by quantifying patterns in the maturity ogive and size- and age-dependent fecundity.
- * Assess whether density dependent shifts in age- or condition-dependent timing of age at maturity have occurred as in other sciaenids.
- * Assess whether temporal patterns in recruitment slope or asymptote have occurred.
- There are no standard protocols for ageing of Atlantic croaker.
- * Conduct a workshop to develop and approve ageing standards for Atlantic croaker.

- * Continue collection of coast-wide age samples from fisheries-independent surveys and length samples from the MRFSS.
- Selectivity curves were used for both commercial and fisheries-independent indices.
- * Evaluate culling of the larger fish out of the survey indices to better match the assumed selectivity.